Hieu Vo

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Education

Villanova University Expected May 2028

B.S. in Computer Engineering, Minor in Computer Science and Artificial Intelligence *GPA: 3.6/4.0*

- **Relevant Coursework:** Data Structures & Algorithms, Web & Mobile Dev, Database Systems, Computer Architecture, Operating Systems, Applied Machine Learning, Computer Networks, Network Security, Signal Processing
- **Honors/Awards:** Dean's List (All Terms), RevolutionUC Hackathon Winner – Best Use of MongoDB Atlas

Experience

Sharedi.io May 2026 – Current

Software Engineer Intern *Mountain View, CA*

- Architected a distributed notification engine using Java, Spring Boot, Kafka, and CockroachDB, supporting **100K concurrent users** across independently scalable services.
- Built Kafka event pipelines processing **5K events per sec**, leveraging priority queues and parallel consumers to distribute notifications across email, SMS, and in-app channels for up to **500 recipients per event**.
- Engineered a distributed Redis caching and Pub/Sub layer, reducing unread notifications count lookup latency by **90%** and delivering **95%** of notifications within **200 ms** via WebSocket on Compute Engine.
- Deployed Docker-containerized microservices on Google Cloud App Engine, recovering from crashes in **under 30 seconds** and preventing **message loss and duplicates** through Kafka consumer replay and idempotent processing.

Media Landscape (MeLa) Lab Oct. 2025 – May 2026

Software Engineer Intern *Villanova, PA*

- Built a Django REST and PostgreSQL analytics platform over **40M** news articles, enabling researchers to compare media bias across **500+ outlets**; optimized database indexes to reduce query latency from **2.5 to 1.5 seconds**.
- Implemented a Kafka-based NLP pipeline with parallel Python workers, increasing ingestion from **15K to 60K articles/day** and reducing batch processing time from **5 hours to 1 hour**.
- Developed 10 automated Python data pipelines to process **60K articles daily** from 500 outlets, transforming raw article data into structured features exposed through 10 REST API endpoints.
- Built a BERTopic-based processing pipeline over **50K articles** to identify major news themes and support automated topic assignment across a **40M-article corpus** for downstream analytics.

L3Harris Technologies Jun. 2025 – Jul. 2025

Software Engineer Intern *Bryn Mawr, PA*

- Engineered a high-throughput image-processing workflow using Python and OpenCV to convert 4K wind tunnel videos into **12K paired training images** for an AI-based airfoil screening system.
- Optimized a GPU-based model training pipeline through adaptive batch sizing and memory management, reducing training time from **10 hours to 2 hours**.
- Developed a containerized aerodynamic screening service using FastAPI, MySQL, and Docker, exposing model predictions through backend APIs for **on-demand airfoil evaluation**.

Projects

AI Coding Memory System | *Go, PostgreSQL, pgx, Docker Compose, golang-migrate*

- Implemented a persistent Go daemon and PostgreSQL backend for cross-session AI coding memory, using pgx connection pooling to serve multiple coding-tool clients without **per-command database initialization**.
- Built PostgreSQL full-text search with weighted indexing and relevance ranking, achieving **sub-second latency** while capping retrieved context at **2,000 tokens** to reduce AI context overhead.
- Reduced local setup to **under 5 minutes** by automating PostgreSQL setup, Docker Compose startup, and database migrations, while keeping the system local with generated credentials and secure config permissions.

Clinical Trial Matching Platform | *Python, FastAPI, MongoDB, Gemini API, LangChain, LangGraph, ChromaDB*

- Architected a clinical trial recruitment platform using Python, FastAPI, MongoDB, and React.js, orchestrating multi-agent AI workflows that reduced patient to trial matching latency from **8 seconds to 4 seconds**.
- Moved OCR processing client-side with Tesseract.js, replacing **5 MB** patient-document uploads with **50 KB** extracted-text payloads while preventing raw documents from reaching the backend.
- Built a structured retrieval and reasoning pipeline using ChromaDB and LangChain, LangGraph with Gemini-powered agents, improving trial-matching accuracy from **65% to 78%** on patient-trial eligibility evaluations.

Technical Skills

Languages: Python, Java, Go, C, C++, JavaScript, TypeScript, SQL
Frameworks & Libraries: Spring Boot, React, Django REST, FastAPI, PyTorch, OpenCV, LangChain, LangGraph
Systems & Databases: PostgreSQL, CockroachDB, MongoDB, Redis, Kafka, RabbitMQ, Docker, GCE, Linux, Git